

Perturbations of Higher Dimensional Black Holes

E645G

1 Introduction

1.1 Gravity in $d = 4$

Black holes (BHs) are the simplest and at the same time some of the most interesting objects in classical general relativity. In our current astrophysical models, BHs are formed during the gravitational collapse of large stars where neither the pressure from nuclear reactions nor degeneracy pressure can support the infalling matter anymore. At the end of this process, the spacetime is a vacuum solution of the Einstein equation (EE)

$$R_{ab} = 0 \quad (1.1)$$

with a curvature singularity. In $d = 4$ dimensions, there exists a uniqueness theorem due to Carter and Robinson [1] that

Theorem 1.1. *If $(\mathcal{M}, g)$ is a stationary, axisymmetric, asymptotically flat, 4-dimensional vacuum spacetime suitably regular on, and outside, a connected event horizon then $(\mathcal{M}, g)$ is a member of the Kerr family of solutions parametrised by the mass M and angular momentum J .*

Hence the most general stationary BH in $d = 4$ is Kerr solution which can be written as

$$ds^2 = -dt^2 + \frac{\mu}{r\Sigma}(dt - a \sin^2(\theta)d\phi)^2 + \frac{\Sigma}{\Delta_4}dr^2 + \Sigma d\theta^2 + (r^2 + a^2) \sin^2(\theta)d\phi^2 \quad (1.2)$$

where $\Sigma = r^2 + a^2 \cos^2 \theta$, $\Delta_4 = r^2 + a^2 - 2Mr$ and $\mu = 2M$ with $a = J/M$. In (1.2) we note that the outer event horizon, at the larger of the roots $\Delta_4(r_+) = 0$, always has a topology S^2 .

The requirement of existence of such roots of $\Delta_4(r)$ prescribes a maximum angular momentum $|a| = M$ known as an extremal bound. This maximum angular momentum naturally exists as for any higher angular momenta, gravity is not able to bind together the matter of a star and collapse would be prevented by centrifugal forces. There is an ongoing debate on whether exactly extremal Kerr BHs are able to form [2].

1.2 Gravity in $d \geq 5$

This could be the end of the story but there is some motivation to investigate general relativity and particularly BHs in dimensions $d \geq 5$.

One reason stems from string theory, a proposal for a quantum gravity framework. String theory naturally lives in ten or more dimensions. To match this with our observation that our universe appears to have four dimensions, the additional dimensions are compactified on a small scale, for example over a Calabi-Yau manifold [3]. As in the $d = 4$ case, in higher dimensional gravity BHs are physically unavoidable and hence it is of interest to study them in this setting to probe the properties of gravity and its role in string theory. In fact, a derivation of the BH entropy from counting microstates has been achieved in string theory which makes this application very promising. [4]

Another more directly motivated reason is the gauge-gravity duality, first proposed by Juan Maldacena in [5], relating certain conformal field theories in d dimensions to gravity in Anti-deSitter (AdS) spacetime in $d + 1$ dimensions. BHs are especially useful in this regard as they map to quantum field theories at finite temperature. These strongly coupled gauge theories are notoriously difficult to study, as perturbative methods fail, and hence changing perspective to a classical BH problem is desirable. We consider a BH in AdS spacetime which has a timelike boundary. This is dual to a gauge theory living on the conformal boundary of the spacetime where the mapping is performed through a dictionary that relates the partition function of the gauge theory and gravitational partition function of which BHs are often the dominant saddle point. A nice method for reconstructing BH images of a spacetime dual to a gauge theory using the behaviour of null

trajectories given data on the boundary is discussed in [6]. This means that studying the classical properties of BHs in $d \geq 5$ can give information about the behaviour of quantum theories at finite temperature. [7] A final motivation comes from the nature of our description of gravity itself. In statistical physics, making d a parameter allows understanding theories such as the Ising model at a much more fundamental level by analysing how phases and critical points change across dimensions. For gravity, in $d \leq 3$ there are no asymptotically flat BHs at all. Our $d = 4$ spacetime allows the existence of BHs and other interesting phenomena. This could indicate that we sit at a critical dimension where the nature of gravity changes. To find out whether $d = 4$ is in fact special, we have to study gravity in $d \geq 5$.

Gravity in $d \geq 5$ is more interesting than in $d = 4$ as the uniqueness theorem 1.1 does not hold anymore. This becomes particularly apparent when considering features of BHs in higher dimensions that do not appear in $d = 4$.

The first is extended black objects. For a metric in block-diagonal form where the components in each block depend only on coordinates corresponding to it, the EE decouples for different blocks. Hence we can add flat directions to the metric to form black strings (BSs) and black branes (BBs) which are guaranteed to be solutions. These extended objects do not appear in $d = 4$ as there are no BHs in $d < 4$ that could be extended. We discuss extended black objects in section 2. [8]

The second is multiple independent angular momenta. In $d = 4$, angular momentum is the generator of the group of spatial rotations $SO(3)$ which has rank 1. This means that there are no independent generators in the Lie algebra $\mathfrak{so}(3)$ and only one angular momentum to fully specify the rotation of any object. In higher dimensions, we replace $SO(3)$ with the spatial rotation group $SO(d-1)$ which has $\text{rank}(\mathfrak{so}(d-1)) = N = \lfloor \frac{d-1}{2} \rfloor$ and so N independent planes of rotation. For $d = 5$ and $d = 6$ we have $\mathfrak{so}(4)$ and $\mathfrak{so}(5)$ respectively which have rank 2 and hence we can specify rotation in two independent planes. We discuss this in detail in section 3. [8]

Finally, for spherical BHs with horizon topology S^{d-2} , in the Newtonian limit the gravitational field is expected to drop off as $\sim 1/r^{d-3}$ as it spreads out over all dimensions, whereas the centrifugal effective potential for a rotating object goes as $\sim 1/r^2$ due to rotation being confined to a plane. This means that in $d \geq 5$, it is possible for an extended, curved horizon to be stabilised against collapse by angular momentum which gives rise to black rings (BRs) with horizon topology $S^{d-3} \times S^1$. We discuss BRs in section 4. [9]

As $d \rightarrow \infty$, the gravitational field drops off almost instantly and all interesting dynamics are confined close to the horizon. General relativity then simplifies dramatically so considering this limit will be of interest here. We discuss this in multiple contexts. [10]

In higher dimensions, there is a zoo of BH solutions, most of which have not yet been classified. Non-uniqueness also allows the dynamics of BHs in higher dimensions to be richer than what is found in $d = 4$. Results indicate that BHs in higher dimensions can be unstable to perturbations and transition to other types of BHs. This is the main topic of this essay.

1.3 Quasinormal Modes

When a BH is formed either from the collapse of a star or the merger of two smaller BHs, it is not exactly a stationary Kerr solution but rather some small perturbation h_{ab} of it [11]. We define small in the sense that the components of the perturbation and their derivatives are small compared to that of the background metric in any given basis. The Ricci tensor receives a perturbation

$$R_{ab} \rightarrow R_{ab} - \frac{1}{2} \Delta_L^{(4)} h_{ab} + O(h^2) \quad (1.3)$$

where $\Delta_L^{(4)}$ is a second order linear differential operator [12]. As the whole process is happening in a vacuum, the perturbed Ricci tensor has to satisfy the vacuum EE which gives

$$\Delta_L^{(4)} h_{ab} = 0, \quad (1.4)$$

a second order PDE that can be used to find the time evolution of the perturbation. At late times, perturbations are of the form of quasinormal modes $h_{ab} \sim e^{-i\omega t}$ where ω is a complex number of which the spectrum does not depend on the type of perturbation but only on the BH. [13]

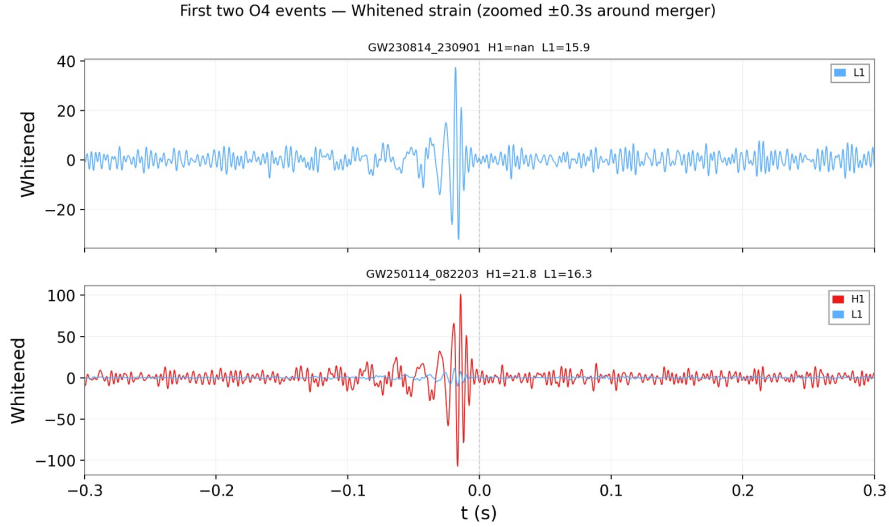


Figure 1: Gravitational waveforms from two BH mergers with ringdown.
Image credit: LIGO collaboration [14], Hamza Mian and John Colling.

It has been attempted to measure this behaviour at LIGO which can be seen from the most recent run in fig. 1. In the first half of the each image, the inspiral gives an oscillating pattern of gravitational waves (GWs). Then there is a peak at the point of merger around $t \approx -0.02$ s and a decay back to background noise levels through the ringdown, a superposition of quasinormal modes. It has not yet been possible to resolve multiple cycles of the ringdown. If successful in the future, this would go a long way to providing observational evidence for the Kerr solution which is used to predict the frequencies.

For a Kerr BH, it can be shown mathematically that all modes have a frequency ω with negative imaginary part. This shows that BHs in $d = 4$ are dynamically stable to linear order although a rigorous proof of full non-linear stability is not available yet. [15]

In this essay, we investigate the stability of higher-dimensional BHs to perturbations. Our central questions are when instabilities can appear and their implications for the predictability of general relativity in higher dimensions. On the way, we review different methods for diagnosing instabilities, comparing their effectiveness and reliability.

2 Black Strings and Branes

2.1 Geometries

We first investigate non-rotating BHs. The Schwarzschild metric for a non-rotating BH in $d = 4$ is

$$ds^2 = - \left(1 - \frac{2M}{r} \right) dt^2 + \frac{dr^2}{1 - \frac{2M}{r}} + r^2 d\Omega_2^2 \quad (2.1)$$

with a horizon topology of S^2 . This can be generalised to the Schwarzschild-Tangherlini solution in $d \geq 5$ as

$$ds^2 = - \left(1 - \frac{\mu}{\tilde{r}^{d-3}} \right) dt^2 + \frac{d\tilde{r}^2}{1 - \frac{\mu}{\tilde{r}^{d-3}}} + \tilde{r}^2 d\Omega_{d-2}^2 \quad (2.2)$$

with a horizon topology S^{d-2} where $\mu \sim M$ is the only parameter. In the Newtonian limit, we obtain the expected drop-off of the potential as $\sim 1/\tilde{r}^{d-3}$. Adding a flat dimension to (2.1) gives BSs ($d = 5$) and BBs ($d \geq 6$)

$$ds^2 = - \left(1 - \frac{r_+}{r} \right) dt^2 + \frac{dr^2}{1 - \frac{r_+}{r}} + r^2 d\Omega_2^2 + \sum_{i=1}^{d-4} dz_i^2 \quad (2.3)$$

with a horizon topology $S^2 \times \mathbb{R}^{d-4}$ where $r_+ = 2M$ is the Schwarzschild event horizon. Due to the infinite length, this BH will have an infinite mass. To avoid this, we compactify the extra flat dimensions to a torus. This gives the horizon topology $S^2 \times \mathbb{T}^{d-4}$ where the circles are not contractible as they are stabilised by topology. [8]

Taking a Schwarzschild-Tangherlini solution in any dimension and adding flat, compactified dimensions to it, another more general non-rotating BH with horizon topology $S^n \times \mathbb{T}^{d-2-n}$ is obtained.

2.2 Gregory-Laflamme Instability

Linear Perturbation Theory

We now consider a BS in $d = 5$ as specified by the metric (2.3) and follow the procedure set out in [12, 16]. The test for instability can be carried out by adding a small perturbation h_{ab} to the metric and solving

$$\Delta_L^{(5)} h_{ab} = \square h_{ab} + 2R_{abcd} h^{cd} - 2R_{(a}^c h_{b)c} - 2\nabla_{(a} \nabla^c h_{b)c} + \nabla_a \nabla_b h = 0 \quad (2.4)$$

where R_{abcd} and R_{ab} are the unperturbed Riemann and Ricci tensors and $\Delta_L^{(5)}$ is the perturbation operator in $d = 5$. We understand indices raised and lowered with the background BS metric. This equation is the spin-2 massless wave equation in curved spacetime. We can simplify the equation into the form

$$\square h_{ab} + 2R_{abcd} h^{cd} = 0 \quad (2.5)$$

by fixing transverse-traceless gauge. Now there exist several symmetries that decouple and eliminate part of the equations. Only spherically symmetric perturbations will be considered so the $SO(3)$ symmetry implies the (θ, ϕ) sector of h_{ab} is a multiple of the round S^2 metric $h_{\theta\theta} d\Omega_2^2$ with no other angular cross-terms. This only leaves six other independent components for the (t, r, z) sector. Next, the t and z translation symmetries can be used to factor out $h_{ab} = e^{imz} e^{\Omega t} H_{ab}(r)$ with $m, \Omega \in \mathbb{R}$. Additionally, due to the compactification of the z dimension, the wavenumber m has to obey the quantisation condition $m = \frac{2\pi n}{L}, n \in \mathbb{Z}$ for mathematical consistency. We will see later why there is no $e^{i\omega t}$ oscillatory term. The goal of this analysis

is to find out whether there are any values of m for which $\Omega > 0$ such that we have an indication of an instability.

The procedure for ensuring that any instability is not a pure gauge artefact is outlined in [12]. We assume that this is the case here for all $m \neq 0$ modes.

We now look for solutions that are regular at the event horizon $r \rightarrow r_+$ and exponentially decay as $r \rightarrow \infty$. To see how this is done we follow the procedure introduced in [17]. We first change to spherical gauge where $h_{\theta\theta} = 0$. We can now use the EE as well as the gauge conditions to solve for all remaining degrees of freedom $H_{ab}(r)$ in terms of one master variable $\mathfrak{h} = H_{zz}$. This means the problem has dramatically simplified to a Sturm-Liouville-type eigenvalue ODE for Ω as

$$\frac{d^2 \mathfrak{h}}{dr^2} + P_m(r) \frac{d\mathfrak{h}}{dr} + \left(Q_m(r) - \frac{\Omega^2 r^2}{(r - 2M)^2} \right) \mathfrak{h} = 0 \quad (2.6)$$

where P_m and Q_m are functions, the specific form of which can be found in section 3 of [17]. This is good but we can do even better. We define $\tilde{\mathfrak{h}}(r) = w(r)\mathfrak{h}(r)$ and use the tortoise coordinate $r_* = r + 2M \log(\frac{r}{2M} - 1)$ to obtain

$$-\frac{d^2 \tilde{\mathfrak{h}}}{dr_*^2} + V_m(r_*) \tilde{\mathfrak{h}}(r_*) = -\Omega^2 \tilde{\mathfrak{h}}(r_*) \quad (2.7)$$

where the explicit form for $w(r)$ and $V_m(r_*)$ can be found in section 4 of [17]. This has the form of a Schrödinger equation on the domain $-\infty < r_* < \infty$ with a self-adjoint Hamiltonian operator $\mathcal{H} = -\partial_{r_*}^2 + V(r_*)$. Therefore the eigenvalues Ω^2 must be real and so Ω can either be real or imaginary. As we do not care about the purely imaginary case, we neglect this and require Ω to be real.

As a result of spectral theory, imposing boundary conditions of regularity at the event horizon $r_* \rightarrow -\infty$ and exponential decay at spacelike infinity $r_* \rightarrow \infty$ now requires the spectrum of $\mathcal{H}$ to be discrete and hence Ω is quantised. In fact, as is noted in [16] and explored in more detail in [17], the potential of $\mathcal{H}$ only permits one bound state and hence only one negative eigenvalue. Therefore there is at most one real Ω for each value of m .

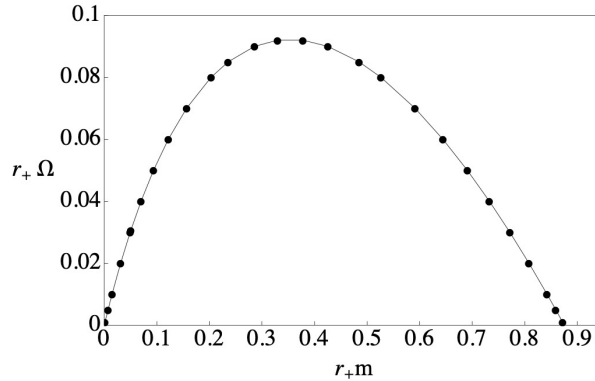


Figure 2: Plot of growth constant with z-wavenumber for perturbed BS.
Image credit: [16]

The resulting eigenvalues for pairs of (m, Ω) are shown in fig. 2. The radial behaviour of the perturbation eigenfunctions can be found in [16]. We see that there is a range of m values for which Ω is positive and hence we have evidence for an instability at linear order.

We would now like to find out what happens to the event horizon. The first issue is that the background BS

metric (2.3) is singular at $r = r_+$. This is easily solved by transforming to Kruskal coordinates $(t, r) \rightarrow (T, R)$ where the metric is analytic and non-degenerate across the event horizon. Another difficulty is that the event horizon is defined as the boundary of the region that cannot signal to future null infinity. As this definition is not local, we cannot find the position of the event horizon without knowing the full evolution of the spacetime to late times. To still get an idea of the position of the event horizon, many numerical relativity computations instead track the apparent horizon, the outermost trapped surface, guaranteed to be contained within the event horizon. We will see this applied later but for now we choose another method, as carried out in [16]. To do this, we note that the event horizon is a null hypersurface generated by null geodesics. These null geodesics can enter but not leave the event horizon. Hence we can use their trajectory to track the position of the event horizon to first order. This is described by the geodesic equation in the perturbed spacetime

$$\left(\frac{dR}{dT}\right)^2 = 1 + \epsilon \cos(mz)(R+T)^{2r_+ \Omega - 2} \left(1 + \frac{dR}{dT}\right)^2 \quad (2.8)$$

where ϵ is a small parameter controlling the magnitude of the initial perturbation. We see that for $\epsilon = 0$ the null geodesics are $R = T + R_0$ and the unperturbed event horizon is at $R = T$. Solving the equation to first order in ϵ , we obtain

$$R = T + \tilde{\epsilon} \cos(mz) T^{2r_+ \Omega - 1} \quad (2.9)$$

which we can transform back into Schwarzschild coordinates to find the radius of the perturbed event horizon as

$$r_{EH}(z) = r_+ + \tilde{\epsilon} T^{2\Omega} \cos(mz). \quad (2.10)$$

We see that the horizon develops ripples along the length of the string.

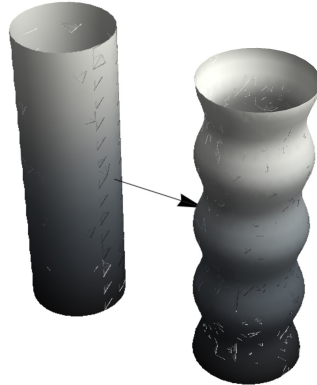


Figure 3: Ripples forming in the event horizon of perturbed spacetime.
Image credit: [16]

This is shown in fig. 3 with the t and θ directions suppressed. Based on (2.10), it would now be tempting to say that as T increases, we eventually reach $r_{EH} \rightarrow 0$ at points along the BS where $\cos(mz) = -1$. This would mean that the event horizon pinches off on the singularity and the BS could split into caged Schwarzschild-Tangherlini-like BHs in the compactified space.

However, we recall that these results were derived to linear order perturbation in h_{ab} so (2.10) can only be valid as long as $T^{2\Omega} \ll r_+$. Therefore, we need a different method to gather evidence whether this pinching off is a realistic end point of the growing modes.

Thermodynamic Argument

To do this, we use BH mechanics, following the approach of [12, 16]. BH mechanics proposes that BHs act as thermodynamic objects with certain identifications made. The second law of BH mechanics is

Theorem 2.1. *Let $(\mathcal{M}, g)$ be a strongly asymptotically predictable spacetime satisfying the Einstein equation with the null energy condition. Let $U \cap \mathcal{M}$ be a globally hyperbolic region for which $J^-(\mathcal{I}^+) \cap U$. Let Σ_1, Σ_2 be spacelike Cauchy surfaces for U with $\Sigma_2 \subset J^+(\Sigma_1)$. Let $H_i = \mathcal{H}^+ \cap \Sigma_i$. Then $\text{area}(H_2) \geq \text{area}(H_1)$.*

This is equivalent to the second law of thermodynamics if we identify $S = \frac{k_B c^3 A}{4G\hbar}$ where $A = \text{area}(\mathcal{H}^+ \cap \Sigma)$ is the area of the event horizon and the total entropy S is non-decreasing. We note this theorem is only true classically. Adding quantum fields, the BH can evaporate through Hawking radiation, decreasing the horizon area. The second law of thermodynamics is not violated as the radiation carries entropy. Before we go to natural units in the expression for S , we note that the value of Newton's constant G_4 in $d = 4$ differs from G_d in a general dimension d by a factor of the compactification volume V_{d-4} . In our case this is $V_{d-4} = L$ which gives $G_5 = G_4 L$. We can now set G_4 and the other constants equal to 1 in natural units. Computing the event horizon areas for the BS and a BH with equal mass in the compactified space we have

$$S_{BS} = 4\pi M^2 \sim M^2, S_{BH} = 4\pi M^2 \sqrt{\frac{8L}{27\pi M}} \sim M^{\frac{3}{2}} L^{\frac{1}{2}}. \quad (2.11)$$

By the second law, the system will tend to maximise its entropy. Therefore, the black object with the higher entropy is expected to be preferred and the other a meta-stable state. We can see that either decreasing the mass or increasing the length scale of the compactification leads to the BH eventually being preferred. This indicates that the pinch-off and then fragmentation into spherical BHs may be viable.

General relativity and fluid dynamics share many deep links. We can see the first hints of this here. In $d \geq 5$, the gravitational field drops off more quickly with distance and event horizons are more flexible, behaving like the boundary of a fluid. This means that we are able to use results from fluids to gain insight on the dynamics of horizons or even treat BHs using effective fluid descriptions in some situations. A jet of fluid can break up into droplets when it becomes too thin and droplets are thermodynamically preferred. This encodes a mathematically equivalent phenomenon to the Gregory-Laflamme instability. [18]

Failure of Intuition

The thermodynamic argument, although very powerful, has flaws. One issue is that we do not compute the entropy at every point and so we do not know whether there is some barrier to the process happening. Even if one of the black objects is meta-stable, we are not guaranteed to evolve there. Quantum mechanically, a system will always eventually tunnel to the most stable state but this is not true classically. One deciding factor here is the order of the phase transition. For first order phase transitions, we have a discrete jump in some order parameter and hence there could be a barrier that forbids the process. For higher order transitions, there is no such discrete jump and we may trust the global thermodynamics more. Further, we do not know if the BH is a global maximum of entropy or whether there are more stable configurations, such as a non-uniform object, that the system can evolve to.

To illustrate this point, we consider a result obtained by [19]. We start with a BB as specified in (2.3) for $d = 6$ where the additional dimensions x_1, x_2 are compactified to an oblique torus $x_1 \sim x_1 + (n_1 + \gamma n_2)L, x_2 \sim x_2 + n_2 \sqrt{1 - \gamma^2}L$ for any $n_1, n_2 \in \mathbb{Z}$ with parameters L and $\gamma = \sin(\theta) \in [0, 1)$ specifying the compactification. We consider perturbations of the form $h_{ab} \sim e^{i\omega t + im_1 x_1 + im_2 x_2}$ where both

m_1 and m_2 satisfy appropriate quantisation conditions.

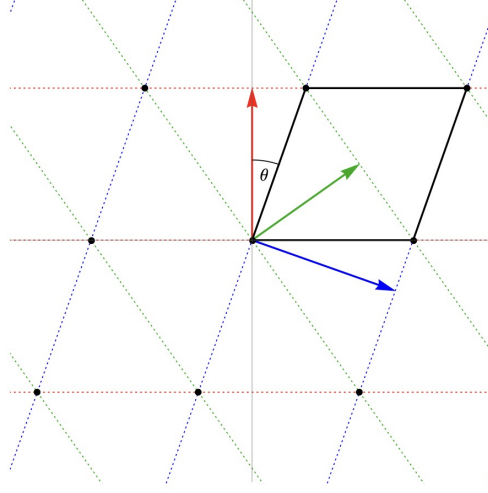


Figure 4: Torus compactification as lattice with fundamental torus region in solid black parallelogram. Allowed wave vectors form sets of parallel lines. Shortest wave vectors are indicated with arrows.
Image credit: [19]

Fig. 4 shows the torus compactification as a lattice where the fundamental region is the solid black parallelogram with opposite sides identified. The wave vectors for the plane waves with the longest allowed wavelengths are indicated as arrows. We can now define a new coordinate system $x_1 = \frac{L}{2\pi}(x + \gamma y)$, $x_2 = \frac{L}{2\pi}\sqrt{1 - \gamma^2}y$ where the compactification identifies $x \sim x + 2\pi$, $y \sim y + 2\pi$ on a square lattice. The allowed wave vectors are

$$|m| = \sqrt{m_1^2 + m_2^2} = \frac{1}{\lambda} \sqrt{n_x^2 + n_y^2 - 2n_x n_y \gamma} \quad (2.12)$$

for any $n_x, n_y \in \mathbb{Z}$ where $\lambda = \frac{\sqrt{1 - \gamma^2}}{2\pi} L$ is the length scale of compactification.

When L/r_+ is small, we do not expect an instability in the uniform BB. For an instability to appear, there has to be a long-wavelength mode with $|m| < m_{GL}$ for the threshold value m_{GL} beyond which modes become unstable. As L is increased, the modes that become unstable first are long wavelength modes with the smallest $|m|$. These modes are

$$n_x = \pm 1, n_y = 0 \text{ and } n_x = 0, n_y = \pm 1 \text{ with } |m| = 1/\lambda \quad (2.13)$$

as well as

$$n_x = n_y = \pm 1 \text{ with } |m| = 2(1 - \gamma)/\lambda. \quad (2.14)$$

The wavelengths are equivalent when $\gamma = \frac{1}{2}$. Any perturbations with shorter wavelengths only become unstable for larger L .

Now it is beneficial to distinguish between three types of BBs. We have already met *uniform* BBs in (2.3). A BB perturbed with one wave vector looks like the product of a non-uniform BS in $d = 5$ with an extra flat direction which is called a *semi non-uniform* BB. These objects do not have a higher entropy than uniform BBs in $d = 6$. A BB with multiple perturbations generated by different wave vectors is called a *fully non-uniform* BB. These will be the objects of interest here.

We can now consider an ansatz metric for the fully non-uniform BB with a parameter ε that controls how non-uniform the BB is. This allows perturbing around the uniform solution at $\varepsilon = 0$ by varying the parameter a small amount. The study in [19] found that, when $\gamma = 1/2$, entropy scales as $\Delta S \sim \varepsilon^3$ and hence increases when $\varepsilon > 0$. This means non-uniform BBs are a possible end point to the Gregory-Laflamme instability in this case.

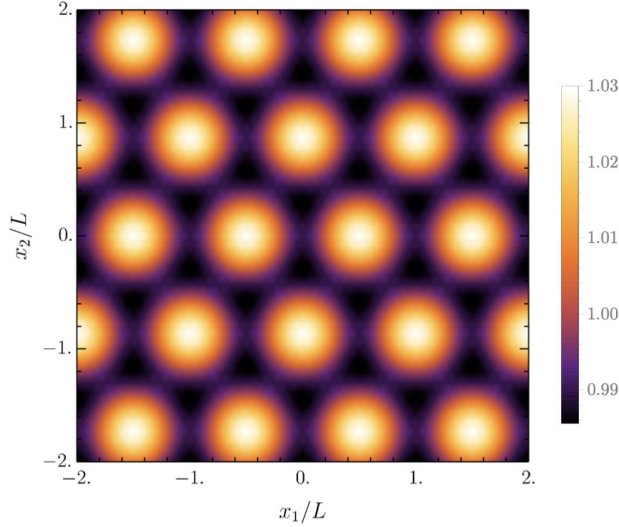


Figure 5: Structure of non-uniform BB at $\gamma = \frac{1}{2}$ where colour indicates event horizon radius.
Image credit: [19]

An example of such a higher-entropy configuration is shown in fig. 5. It is unknown whether this can further develop into multiple localised BH with horizon S^4 but a mechanism has been suggested where the BB breaks up into BHs connected by BSs which then break by a Gregory-Laflamme instability.

As an aside, there is a fascinating feature of the configuration found in [19]. The arrangement of BHs on the lattice with the highest entropy in fig. 5 corresponds to the a triangular lattice which is the densest sphere packing in two dimensions with one BH per torus volume. This leads [19] to conjecture that there is a link between the geometric problem of dense sphere packing and entropically preferred configurations.

Another example of a viable end point without horizon topology change is BSs, generalised to a $d - 1$ dimensional Schwarzschild-Tangherlini BH with a flat direction z as done by [20]. As usual, we compactify the flat direction to $z \sim z + L$. The approach for constructing non-uniform BSs here is similar to the one for constructing non-uniform BBs. We consider an ansatz metric and then perturb around the uniform solution. This allows for the computation of the entropy of non-uniform BSs which can be compared with that of uniform BSs. It is found in [20] that in $d = 12, 13$, there exist highly non-uniform configurations which appear stable and in $d = 14, 15$ all BSs have non-uniform configurations as viable end points. A possible explanation for this is argued by [21] which suggests that there is a critical dimension $d_* = 13$ for the uniform/non-uniform phase transition where for $d \leq d_*$, the phase transition is first order, tending to fragmentation into BHs and for $d > d_*$ the transition is of a higher order, tending towards the formation of non-uniform BSs. Above d_* , the non-uniform BS has a higher entropy than the uniform BS and hence it is a plausible end point.

At large d , this end point can be confirmed by [22]. To do this, they consider the ansatz metric of a boosted BS with variables $\mathfrak{m} = (r_{EH}(z))^{d-4}$, the mass density, and $\mathfrak{p}(z)$, the momentum density along the string.

As $d \rightarrow \infty$, the EE can be written in the simplified form

$$\partial_t \mathfrak{m} - \partial_z^2 \mathfrak{m} = -\partial_z \mathfrak{p}, \quad \partial_t \mathfrak{p} - \partial_z^2 \mathfrak{p} = \partial_z \left(\mathfrak{m} - \frac{\mathfrak{p}^2}{\mathfrak{m}} \right) \quad (2.15)$$

which looks like fluid equations in one dimension. These equations can easily be solved numerically for small perturbations around the uniform solution. Hence [22] show that for large enough d , non-uniform BSs are the generic end point to the Gregory-Laflamme instability.

Numerical Evolution

To gain more clarity on how the BS in $d = 5$ evolves with the Gregory-Laflamme instability, [18] perform a numerical evolution of the initial data, keeping track of the position of the apparent horizon.

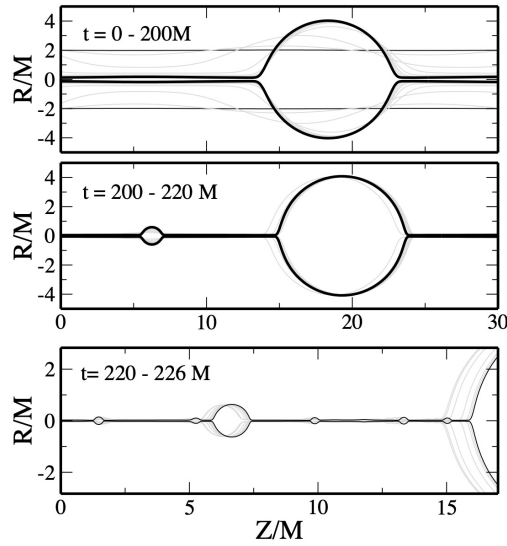


Figure 6: Position of apparent horizon in numerical simulation of Gregory-Laflamme instability.
Image credit: [18]

In fig. 6, it can be seen that the apparent horizon shows that the BS develops into localised BHs connected by segments of thin BSs like pearls on a necklace. The pattern then repeats in a self-similar way with the segments developing their own Gregory-Laflamme instabilities and so on. Although the resolution of the simulation is finite, the ratio of the self-similar cascade can be used to estimate the time when the event horizon pinches off. This can be shown to happen in a finite time.

An issue with this is that only the position of the apparent horizon can be studied rather than the event horizon. Mathematically, it is guaranteed that the apparent horizon is inside the event horizon but not that it is necessarily close to it. As an extreme example, there exists a particular slicing of Schwarzschild spacetime where the apparent horizon is arbitrarily close to the singularity. In the worst case the simulation predicts a pinch-off in the horizon where there is none. [23]

A more recent analysis in [24] confirms the result of [18] with the further suggestion that, based on a comparison with fluid dynamics, the exact dynamics of the pinch-off at the microscopic level, where the event horizon and apparent horizon may differ, do not affect the macroscopic evolution.

Comparison of Approaches

We have now seen that none of the tools alone are perfect to decide the stability of BSs and BBs. Perturbation theory shows the existence of unstable modes that demonstrates the onset of the instability but breaks down as the instability grows exponentially and out of the perturbative regime.

Thermodynamics is important in determining the viability of a proposed end point of the instability but is not sensitive to the intermediate dynamics or alternative end points. Further, while the second law 2.1 gives a necessary condition for a system to develop into a certain direction, it is not sufficient classically. There is no guarantee whether we will evolve towards the maximum entropy state or just some higher entropy state. For BSs and BBs, a lot of evidence points towards the Gubser-Mitra conjecture, first introduced in [25] and later discussed in [26], which states that for extended black objects with translational invariance, thermodynamic stability is equivalent to classical stability. This suggests that we can trust the thermodynamic argument here.

Numerical evolution is the most powerful tool, giving information about full process but is computationally expensive, has a finite resolution and can be dependent on the spacetime slicing. Taken together, these methods give strong evidence that the Gregory-Laflamme instability leads to a bifurcation of the BS into BHs in a finite time. This allows us to propose a phase diagram for non-rotating BHs in these type of compactified spaces. At large mass the extended horizon is preferred but as the mass decreases the compact BH becomes more favourable.

One argument against this is that, generically, any black object rotates. Therefore it is interesting to see whether rotation in the spherical directions can remove the Gregory-Laflamme instability. It is shown in [27] that this does not happen and in fact the Gregory-Laflamme instability is enhanced. Additionally, as shown in [28] and [29], rotation also introduces new instabilities that are not present for non-rotating BSs. We will analyse some of them when going on to rotating BHs.

Another method we could use to attempt to remove the instability generically is giving BSs and BBs an electromagnetic charge. For a small charge, as discussed in [30], it is shown that the instability still persists. Hence the instability is also generic in this regard. Interestingly, as we increase the charge to the near extremal limit, [31] are able to show that the instability seems to disappear. The authors of [31] go on to argue that this is expected as the entropy of a BH goes to zero more quickly than that of a BB and so the transition becomes forbidden by the second law. It is important to mention that, while the result of the analysis in [31] is still valid, it is not believed anymore that extremally-charged BHs have zero entropy as a result of the work [4] and others. Furthermore, extremal BBs as considered in [31] are supersymmetric, saturating a BPS bound, and should hence be stable as they minimise energy. However, as this is not the generic case, it is not expected to hinder the mechanism of the Gregory-Laflamme instability in general.

2.3 Cosmic Censorship

For a classical theory, we wish that, given a reasonable data set, we are able to use the equations of motion to predict the fate of all observers for all time. To this end we define [1]

Definition 2.1. *An asymptotically flat spacetime $(\mathcal{M}, g)$ is strongly asymptotically predictable if there exists an open region $\bar{\mathcal{V}} \subset \bar{\mathcal{M}}$ such that $\mathcal{M} \cap \bar{\mathcal{J}}^-(\mathcal{I}^+) \subset \bar{\mathcal{V}}$ and $(\bar{\mathcal{V}}, \bar{g})$ is globally hyperbolic where $(\bar{\mathcal{M}}, \bar{g})$ is an analytic extension of $(\mathcal{M}, g)$.*

There are two conjectures due to Penrose that propose that general relativity is classically predictable. The weak cosmic censorship conjecture (WCCC) states that

Conjecture 1. Let (Σ, h_{ab}, K_{ab}) be a geodesically complete, asymptotically flat, initial data set. Let the matter fields obey hyperbolic equations and satisfy the dominant energy condition. Then generically

the maximal Cauchy development of this initial data is an asymptotically flat spacetime that is strongly asymptotically predictable.

The strong cosmic censorship conjecture (SCCC) states that

Conjecture 2. Let (Σ, h_{ab}, K_{ab}) be a geodesically complete, asymptotically flat, initial data set. Let the matter fields obey hyperbolic equations and satisfy the dominant energy condition. Then generically the maximal Cauchy development of this initial data is inextendible.

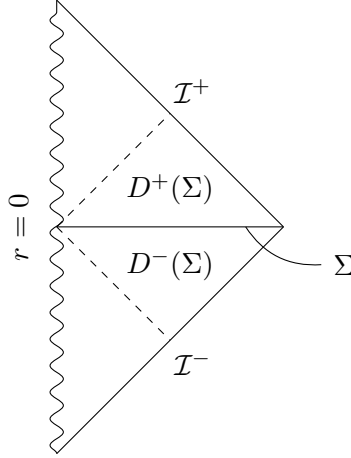


Figure 7: Penrose diagram of an eternal naked singularity with partial Cauchy hypersurface Σ and future/past domain of dependence $D^\pm(\Sigma)$.

We can now consider the Penrose diagram for a naked singularity at $r = 0$ as shown in fig. 7. We introduce a partial Cauchy slice Σ as the spacelike hypersurface $t = 0$ and consider its maximal Cauchy development $D^+(\Sigma) \cap D^-(\Sigma)$ which possesses Cauchy horizons.

Now both past and future null infinities are incomplete so the maximal Cauchy development cannot be asymptotically flat. Therefore, the WCCC is violated. The maximal Cauchy development is also extendible which means that the SCCC is violated. This implies that a naked singularity that persists forever renders spacetime unpredictable. This is because, classically, we have to treat the singularity as a hole in the spacetime and hence we cannot predict the behaviour of anything that can be influenced by the singularity after its trajectory passes through the Cauchy slice. In a spacetime where the singularity is covered by a horizon, this is not an issue as the causal future of the singularity is fully contained within the BH region $J^+(\text{singularity}) \subset \mathcal{B}$. Therefore, naked singularities as shown in the Penrose diagram violate both the WCCC and the SCCC.

The naked singularity that arises from the Gregory-Laflamme instability is not expected to persist forever but only for an instant. [16]

To see why this still violates cosmic censorship, we can draw a spacetime diagram of Minkowski spacetime where we delete a single point as shown in fig. 8. Again, we choose a Cauchy slice Σ as some $t = \text{const.}$ spacelike hypersurface and find the maximal Cauchy development does not include the shaded region. Even with a single point deleted, corresponding to a naked singularity appearing only for an instant, we violate both the WCCC and the SCCC as we do not have a complete null infinity and the maximal Cauchy development is extendible. This means that we are not able to predict the fate of all classical observers.

We are left with the possibility of the singularity dividing before the event horizon pinches off which would prevent a naked singularity from forming. However in this case, there is a theorem that

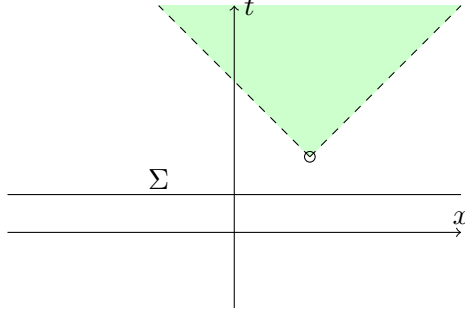


Figure 8: Minkowski spacetime with a single point deleted.

Theorem 2.2. *Let $(\mathcal{M}, g)$ be strongly asymptotically predictable and let Σ_1, Σ_2 be Cauchy surfaces with $\Sigma_2 \subset I^+(\Sigma_1)$. Let B be a connected component of $\mathcal{B} \cap \Sigma_1$. Then $J^+(\Sigma_1) \cap \Sigma_2$ is contained within a connected component of $\mathcal{B} \cap \Sigma_2$.*

The event horizon is forbidden from bifurcating without forming a naked singularity in the process. In fact, BHs are required to pass through a naked singularity whenever changing the topology of the event horizon in any way. Therefore, if we accept the end point of the Gregory-Laflamme instability, then cosmic censorship does not hold in $d \geq 5$. While there is some ongoing debate about whether the violation of cosmic censorship requires finely tuned initial data, the most common position is that this result is generic. [32, 33]

This means that d is not just a quantitative parameter to change in equations but gravity in $d \geq 5$ appears to be fundamentally and qualitatively different to gravity in $d = 4$ as it does not appear to be classically predictable from a reasonable initial data set. Quantum gravity may give more insight into this.

3 Myers-Perry Black Holes

3.1 Geometries and Phase Space

All BHs we considered in the previous section were non-rotating, however we know that the most general BH in $d = 4$ is a rotating Kerr solution as given by (1.2).

The generalisation of this solution to $d \geq 5$ is

$$ds^2 = -dt^2 + \frac{\mu}{r^{d-5}\Sigma} (dt - a \sin^2(\theta) d\phi)^2 + \frac{\Sigma}{\Delta_d} dr^2 + \Sigma d\theta^2 + (r^2 + a^2) \sin^2(\theta) d\phi^2 + r^2 \cos^2(\theta) d\Omega_{d-4}^2 \quad (3.1)$$

where the most important modification is $\Delta_4 \rightarrow \Delta_d = r^2 + a^2 - \frac{\mu}{r^{d-5}}$ with μ as the mass parameter.

We note that there is an event horizon at $\Delta_d(r_+) = 0$ which always has a positive solution for $d \geq 6$. This means that in high dimensions, arbitrarily large angular momenta are allowed while still having a regular event horizon. This is because gravity scales as $\sim 1/r^{d-3}$ while the centrifugal barrier scales as $\sim 1/r^2$ so if $d \geq 6$, gravitational collapse can occur for high angular momenta. This type of BH is known as a singly spinning Myers-Perry (MP) BH.

As mentioned above, in $d \geq 5$ we can also have multiple angular momenta specifying the rotation of an object. In $d = 4$, this rotation takes place in three spatial dimensions and hence we can fully specify it with one plane of rotation and magnitude. Mathematically, angular momentum corresponds to the generators of the spatial rotation group $SO(3)$. The Lie algebra $\mathfrak{so}(3)$ is generated by antisymmetric matrices $(M_{ij})_{kl}$ where the pair (i, j) specifies the plane of rotation giving three generators M_{12}, M_{23}, M_{31} . If the generator

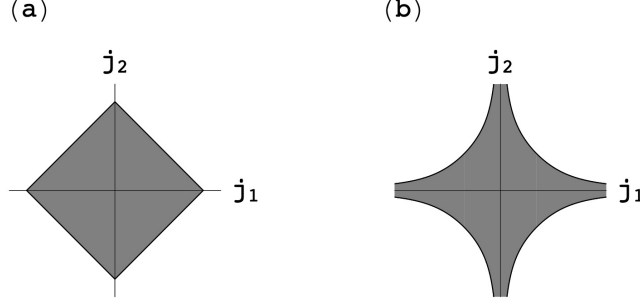


Figure 9: Phase space of MP BH in (a) $d = 5$ and (b) $d = 6$.
Image credit: [8]

matrices share an index (i.e. an axis) they are not independent and hence do not commute. We see that it is not possible to have two commuting generators of $\mathfrak{so}(3)$. Therefore maximal commuting set of generators is 1, giving a Cartan subgroup $U(1)$ and we can only choose angular momentum in one plane. Generalising to d dimensions, we replace $\mathfrak{so}(3)$ with $\mathfrak{so}(d-1)$ which has a maximal commuting set of $N = \lfloor \frac{d-1}{2} \rfloor$ giving a maximal Cartan subgroup $U(1)^N$ and hence angular momentum in N independent planes. To see this, we can consider the set of generators $\{M_{ij}\}$, $1 \leq i, j \leq d-1$ and try to construct sets of independent rotation planes. In $d = 5$ and $d = 6$ for example we get two planes M_{12} and M_{34} . As we always have to pair two different orthogonal axes to obtain a new independent plane of rotation, there are $N = \lfloor \frac{d-1}{2} \rfloor$ for $d-1$ spatial dimensions. [8]

We can now write down the full MP BH solution with arbitrary angular momenta $J_i, i = 1, \dots, N$. The most convenient form to express this in is

$$ds^2 = -dt^2 + (\tilde{r}^2 + a_i^2)(d\mu_i^2 + \mu_i^2 d\phi_i^2) + \frac{\mu}{\Pi F}(dt - a_i \mu_i^2 d\phi)^2 + \frac{\Pi F}{\Pi - \mu r^2} d\tilde{r}^2 \quad (3.2)$$

for odd dimensions and

$$ds^2 = -dt^2 + \tilde{r}^2 d\alpha^2 + (\tilde{r}^2 + a_i^2)(d\mu_i^2 + \mu_i^2 d\phi_i^2) + \frac{\mu}{\Pi F}(dt - a_i \mu_i^2 d\phi)^2 + \frac{\Pi F}{\Pi - \mu r} d\tilde{r}^2 \quad (3.3)$$

for even dimensions. We have $a_i \sim J_i$ at fixed mass and direction cosines $\cos(\theta_i) = \mu_i$. The functions Π and F depend on the coordinates $\tilde{r}, \mu_i$ as well as parametrically on a_i . Their exact form is found in [8]. The coordinate system also requires $\alpha^2 + \mu_i^2 = 1$.

We now have the event horizon at $\Pi(r_+) - \mu r_+^2 = 0$ for odd dimensions and $\Pi(r_+) - \mu r_+ = 0$ for even dimensions. This defines a phase space for the allowed values of $j_i \sim \frac{J_i}{(GM^{d-2})^{1/(d-3)}}$, a dimensionless variable for J_i , required to have a well-defined event horizon. The phase space is summarised for $d = 5, 6$ as the shaded region in fig. 9 where we can see that for a singly spinning MP BH in $d = 6$ the angular momentum can get arbitrarily large. For higher dimensions, there are more angular momenta and hence a higher-dimensional phase space but the shape remains qualitatively the same.

MP BHs with multiple different angular momenta are difficult to analyse as they do not have a large amount of symmetry. For this analysis, we restrict ourselves to singly spinning MP BHs and those with all angular momenta equal $J_i = J$ which are known as cohomogeneity-1 as there is only one free parameter once the mass is fixed.

3.2 Two Types of Instabilities

For MP BHs there are two important types of instabilities.

When the angular momentum of an MP BH becomes very large, the BH is *ultraspinning*. This leads to

the event horizon strongly deforming into a disk which means the BH can be approximated as a BB with horizon topology $\mathbb{R}^2 \times S^{d-4}$. [34]

We already know this can be unstable under a Gregory-Laflamme instability. Therefore, there is a value of a for ultraspinning MP BH in $d \geq 6$ beyond which they become unstable. The end point of the ultraspinning instability has been shown to be a set of concentric BRs, extended black objects similar to BSs but supported by angular momentum instead of topology. We will discuss this object later. [35, 36]

Another type of instability breaks the axisymmetry of a rotating BH with perturbations and the BH is able to divide into multiple BHs of which the orbital angular momentum is equivalent to the original rotational angular momentum.



Figure 10: Numerical evolution of m -bar instability of unstable BH for $m = 4$.
Image credit: [15]

The evolution of such an m -bar instability is shown in fig. 10. It is expected to set in at a lower angular momentum than the ultraspinning instability and lead to extreme GW emission.

As both of these instabilities change the event horizon topology, they are expected to violate cosmic censorship.

Here we can again emphasise the links between general relativity in higher dimensions and fluid dynamics. A quickly spinning droplet of fluid can either turn into a ring or develop a non-axisymmetric instability and split up into smaller droplets.

We will now turn to investigate when these instabilities appear perturbatively.

3.3 Kodama-Ishibashi Formalism

To understand how the perturbation theory is carried out as well as the quantities of interest for these types of BHs, we will first consider the simplest MP BH, a non-rotating Schwarzschild-Tangherlini BH. Due to the high degree of symmetry, there are powerful tools that help simplify the equations to an ODE.

In $d = 4$, the Teukolsky equation simplifies the perturbation equations for the Kerr BH by separating variables into radial and angular contributions which allows them to be solved as a self-adjoint ODE eigenvalue problem.

The Kodama-Ishibashi (KI) formalism seeks to do the a similar decomposition for $d \geq 5$ where we will follow the procedure originally laid out in [37]. This formalism applies to metrics of the form

$$ds^2 = h_{AB}(y)dy^A dy^B + r^2(y)\hat{g}_{ab}dx^a dx^b \quad (3.4)$$

where h_{AB} is a 2-dimensional Lorentzian metric and here $\hat{h}_{ab}$ is a maximally symmetric Riemannian metric of dimension $n = d - 2$. The Schwarzschild-Tangherlini metric is of this type where $\hat{g}_{ab}$ is the round metric for S^{d-2} . Now the time and angular dependence can be factored out as $e^{i\omega t}$ as well as spherical tensor

harmonics on S^{d-2} . These spherical tensor harmonics then decompose into scalar S , transverse vector V_a and transverse-traceless-symmetric tensor T_{ab} degrees of freedom. The scalar harmonics obey

$$(\hat{\nabla}^2 + \lambda_S)S = 0 \quad (3.5)$$

where $\lambda_S = l_S(l_S + n - 1)$ and $l_S = 0, 1, 2, \dots$ is the angular quantum number for the scalar harmonics. $\hat{\nabla}^2$ is the Laplacian on the n -dimensional subspace described by $\hat{g}_{ab}$, in our case S^{d-2} with the round metric. The vector and tensor harmonics satisfy similar equations

$$(\hat{\nabla}^2 + \lambda_V)V_a = 0, (\hat{\nabla}^2 + \lambda_T)T_{ab} = 0 \quad (3.6)$$

where $\lambda_V = l_V(l_V + n - 1) - 1$ and $\lambda_T = l_T(l_T + n - 1) - 2$. Now the time and angular dependence can be factored out and gauge-invariant master variables ϕ_I constructed to obtain the master equation in the form of the radial ODE

$$-f\partial_r(f\partial_r\phi_I) + (V_I - \omega^2)\phi_I = 0 \quad (3.7)$$

where $f(r) = 1 - \frac{\mu}{r^{d-3}}$ and V_I is the effective radial potential which can be found in sections 3 and 5 of [37] where the formalism is first introduced. The index $I = S, V, T$ runs over the scalar, vector and tensor degrees of freedom giving three equations that can be solved using standard methods to find the perturbation modes ϕ_I and frequencies ω .

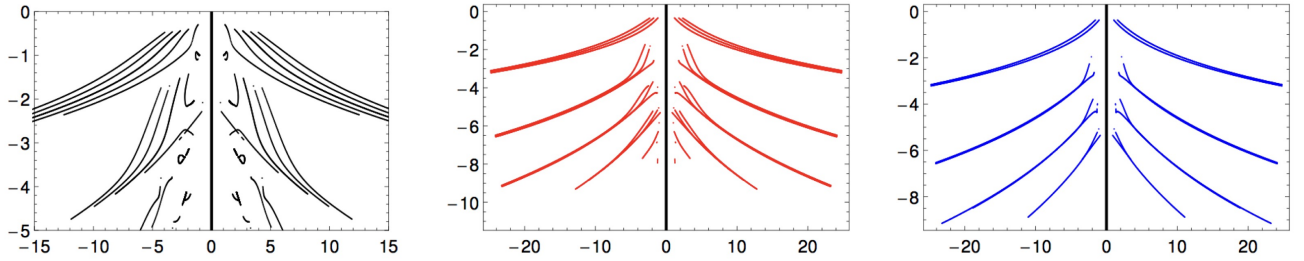


Figure 11: Frequencies of quasinormal modes for a Schwarzschild-Tangherlini BH with scalar (left), vector (middle) and tensor (right) modes. Different curves are for $l_S = 2, 3, 4, 5, 6$, $l_V = 2, 3, 4$ and $l_T = 1, 2$. Dimension is varied from $d = 6$ to $d = 100$ frequencies towards.

Image credit: [11]

These frequencies are plotted in the imaginary plane for different values of angular quantum numbers and dimensions in fig. 11. Due to the $t \rightarrow -t$ symmetry for non-rotating BHs, it can be seen that the modes come in pairs $(\omega, -\omega^*)$. This is no longer the case when we look at rotating BHs. There are no unstable modes as all values of ω have a negative imaginary part. Therefore Schwarzschild-Tangherlini BHs are linearly stable. Nonetheless there are interesting results to be extracted that give us information about how the quasinormal mode frequencies behave as we change angular quantum number and dimension.

For most modes, increasing the dimension increases both the decay rate and the oscillation frequency. As dimension is increased, the frequencies for curves that scale together but have different angular quantum numbers converge to each other. For some scalar and vector perturbations, there are modes that do not scale but rather saturate near their value at a low dimension.

In fig. 12, the behaviour of the scalar quasinormal modes is shown. On the left, the saturating mode starts at the black dot and traces out a shape before converging to the limiting value around $(1.01, 1.00i)$. On the middle and right, the scaling modes have a power-law dependence on the dimension with $\text{Im}(\omega) \sim d^{1/2}$ and $\text{Re}(\omega) \sim d$.

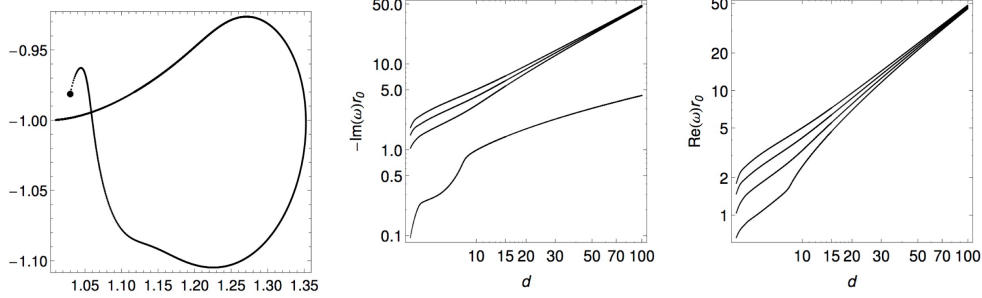


Figure 12: Behaviour of quasinormal mode frequency as d is increased from $d = 6$ to $d = 100$ for a saturating mode for $l_S = 2$ (left) and scaling modes for $l_S = 2, 3, 4, 5$ (middle and right).

Image credit: [11]

For vector modes, the saturating modes are purely imaginary but otherwise there is similar behaviour to the scalar modes. For tensor degrees of freedom, there are no saturating modes.

This leads to interesting behaviour at large d . As in the limit of $d \rightarrow \infty$, scaling modes decay instantly, they can be ignored and the quasinormal mode spectrum simplifies as it is only made of saturating modes. We can now take the insight gained from analysing stable, non-rotating BHs together with the suggested mechanisms of instabilities for rotating BHs to analyse singly spinning and cohomogeneity-1 MP BHs.

3.4 Cohomogeneity-1

The first type of rotating BHs that we consider have cohomogeneity-1 with all angular momenta equal $j_i = j \neq 0$. As these BHs still have a large amount of symmetry, similar techniques as the KI formalism are available. In odd dimensions, the MP metric simplifies drastically as presented in [11] and can be written as

$$ds^2 = -\frac{p(\hat{r})}{h(\hat{r})} dt^2 + \frac{d\hat{r}^2}{p(\hat{r})} + \hat{r}^2 \left[h(\hat{r})^2 (d\psi + A_a dx^a - u(\hat{r}) dt)^2 + \hat{g}_{ab} dx^a dx^b \right] \quad (3.8)$$

where $\hat{g}_{ab}$ is the Fubini-Study metric on $\mathbb{CP}^{(d-3)/2}$. By a process called Hopf fibration, the round sphere S^{d-2} can be described as a principal $U(1)$ bundle over $\mathbb{CP}^{(d-3)/2}$. This works as S^{2n+1} can be identified with the space of non-zero vectors in $\mathbb{C}^{n+1}$ with equivalence relation $(z_1, z_2, \dots, z_{n+1}) \sim (\lambda z_1, \lambda z_2, \dots, \lambda z_{n+1})$, $\lambda \in \mathbb{R}^+$. $\mathbb{CP}^n$ is defined the same way but with $\tilde{\lambda} \in \mathbb{C} \setminus \{0\}$. Therefore S^{2n+1} and $\mathbb{CP}^n$ differ only by a phase which means $\mathbb{CP}^n = S^{2n+1}/U(1)$. Hopf fibration can be written as $U(1) \hookrightarrow S^{2n+1} \rightarrow \mathbb{CP}^n$.

The round metric now decomposes into the Fubini-Study metric as

$$d\Omega_{d-2}^2 = (d\psi + A_a dx^a)^2 + \hat{g}_{ab} dx^a dx^b \quad (3.9)$$

where $\{x^a\}$ are the coordinates on $\mathbb{CP}^{(d-3)/2}$ and ψ is the $U(1)$ phase coordinate. A more detailed explanation on Hopf fibration may be found at [38].

A_a corresponds to the local connection 1-form on the $U(1)$ bundle, similarly to electromagnetism. Considering the isometries of (3.8), the t and ψ directions again factorise as $h_{ab} \sim e^{i(\omega t - m\psi)}$ where m has to be quantised for regularity. We can now perform an SVT decomposition over $\mathbb{CP}^{(d-3)/2}$ where each harmonic also has a charge m associated. The new *charged* harmonics are then eigentensors of

$$\hat{D}_a = \hat{\nabla}_a - im A_a \quad (3.10)$$

where $\hat{\nabla}_a$ is the covariant derivative on $\mathbb{CP}^{(d-3)/2}$. The scalar S , transverse vector V_a and transverse-traceless tensor T_{ab} harmonics now obey equations

$$(\hat{D}^2 + \lambda_I)I = 0 \quad (3.11)$$

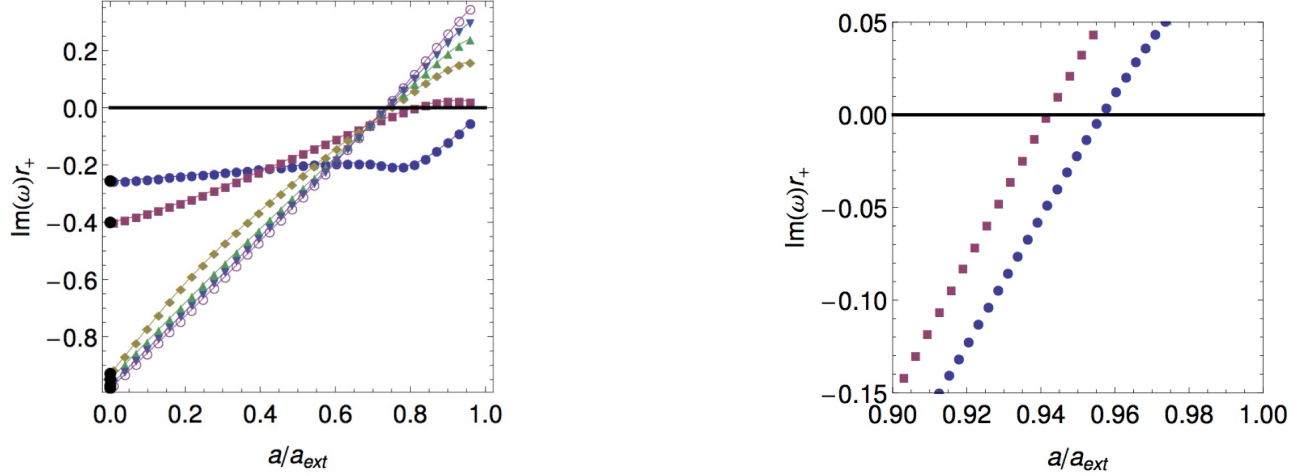


Figure 13: Scalar perturbation frequencies for $(\kappa, m) = (0, 2)$ on left with dimension increasing in odd numbers from $d = 5$ to $d = 15$ from bottom to top at $a = a_{\text{ext}}$. Scalar perturbation frequencies for $(\kappa, m) = (2, 0)$ on right with $d = 11$ as circles and $d = 13$ as squares.

Image credit: [11]

where eigenvalues $\lambda_I, I = S, V_a, T_{ab}$ are functions of two parameters κ and m . The difference to the KI formalism is that here there are no gauge invariant variables so it is common to work in transverse-traceless gauge. The high degree of symmetry again allows the linearised EE equation to be solved as a set of coupled ODEs for a generic perturbation as the metric (3.8) only non-trivially depends on one variable $\hat{r}$. [11]

We now want to find instabilities with $m = 0$ corresponding to an axisymmetric ultraspinning instability and $m \neq 0$ corresponding to a non-axisymmetric m-bar instability.

With all angular momenta equal and non-zero, the values for a are restricted by an extremal limit a_{ext} as can be seen in fig. 9 and hence we may guess that m-bar instabilities are the dominant instability here.

For scalar perturbations, an m-bar instability is found for $\kappa = 0, m \geq 2$ in $d \geq 7$. The left of fig. 13 shows this mode for different dimensions and the full range of allowed angular momenta. It can be seen that $d = 5$ is stable all the way up to the extremal limit and $d \geq 7$ becomes unstable close to a_{ext} . This indicates that higher dimensions are more unstable.

There is also an ultraspinning instability for $\kappa = 2, m = 0$ with $d \geq 7$. The right of fig. 13 shows this for $d = 11$ and $d = 13$. As expected, the ultraspinning instability only occurs at higher a than the m-bar mode instability. We again find that the higher dimension $d = 13$ unstable for a larger range of angular momenta than the lower dimension $d = 11$.

The observation that BHs are more unstable for higher dimensions carries on through all instabilities found. As can be seen particularly from the left image in fig. 13, the frequencies appear to be approaching a limiting value for any given a as $d \rightarrow \infty$. We have already seen this phenomenon for non-rotating BHs in the form of saturating modes. In the limit as $a = 0$, the modes shown on the left in fig. 13 turn out to connect to the $l_S = 2$ saturating mode for $d \geq 9$ and the $l_S = 2$ scaling mode for $d = 5, 7$. This leads [11] to conjecture that for sufficiently high d , all unstable modes connect to saturating modes in the non-rotating limit.

A nice depiction of this can be seen in fig. 14 where the unstable modes connect to the saturating mode for $d \geq 9$. This could be a very important insight as it would imply that at high dimension, all instabilities essentially behave the same. It also indicates that there are unstable, rotating BHs for all sufficiently large

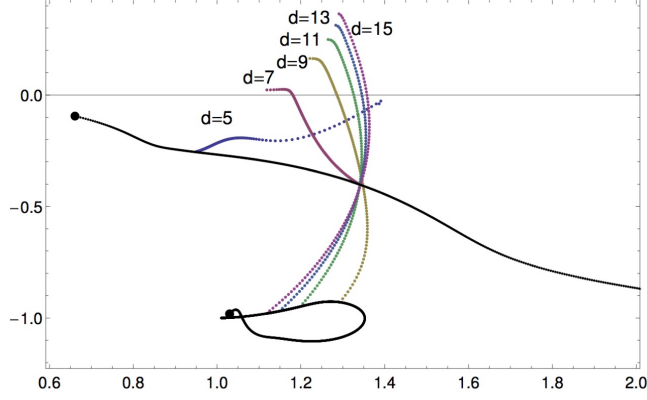


Figure 14: Scalar perturbation frequencies of cohomogeneity-1 for $(\kappa, m) = (0, 2)$ mode for different dimensions and increasing a in colour. Scalar perturbation frequencies of Schwarzschild-Tangherlini for $l_S = 2$ saturating (bottom) and scaling (top) modes for $a = 0$ for increasing dimensions in black.
Image credit: [11]

dimensions, leading to topology change and violation of cosmic censorship. This is a notable difference to BSs and BBs where beyond a certain dimension, it is possible to have configurations where non-uniform strings and BBs are plausible end points. Put together it means that cosmic censorship is violated for any $d \geq 5$.

3.5 Singly Spinning

The singly spinning case also permits a similar SVT decomposition as the non-rotating and cohomogeneity-1 case. The difference to note is that the spherical harmonics are over S^{d-4} instead of S^{d-2} due to the metric being written in terms of a 4-dimensional Lorentzian space and a round S^{d-4} . Solutions are specified by two angular quantum numbers l_S, l_V and m . As there is less symmetry than in the non-rotating and cohomogeneity-1 case, the linearised EE equation is now solved as PDEs in two variables for generic perturbations.

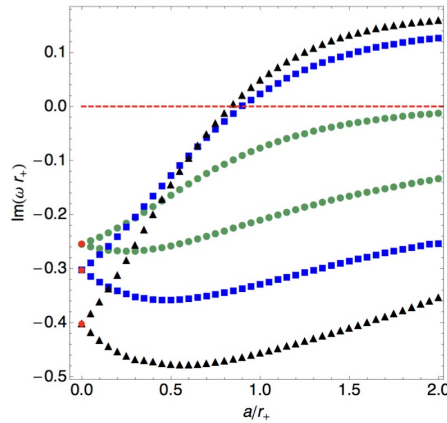


Figure 15: Scalar mode frequencies for $(l_S, m) = (0, 2)$ in dimensions $d = 5$ (green), 6 (blue), 7 (black).
Image credit: [11]

In fig. 15, it can be seen that once rotation is non-zero, there are two scalar modes for each (l_S, m) ,

corresponding to the pairs $(\omega, -\omega^*)$ that have the same imaginary part at $a = 0$ due to the $t \rightarrow -t$ symmetry. For a singly spinning BH, there is a $(t, \phi) \rightarrow (-t, -\phi)$ symmetry instead. This means we can still find pairs of modes $(\omega, -\omega^*)$ but now corresponding to m and $-m$ instead of the same m . This degeneracy is removed when restricting $m \geq 0$.

For $d = 6, 7$, there is a bar mode instability at beyond $a \approx 0.8r_+$. For $d = 5$, it is found that the upper mode does not become unstable but rather exponentially approaches $\text{Im}(\omega) = 0$. Interestingly, the non-linear analysis in [39] finds evidence of a bar mode instability for $d = 5$ here. This tension is resolved by [15] where it is shown that GW emission leads to a loss in angular momentum and the BH settles down to another MP BH before it can become unstable for all values of a up to the extremal bound. A similar process happens for $d \geq 6$ although these are still unstable for sufficiently high spin. Having different tools disagree on stability questions of BHs emphasises how important it is to consider multiple approaches for diagnosing instabilities as well as taking into account the possible mechanisms for instabilities to occur or dissipate.

The vector perturbations are found to be stable and approaching a constant value with $\text{Im}(\omega) < 0$ in the limit as $a \rightarrow \infty$. A similar result holds for tensor perturbations.

At higher angular momenta, [40] was able to show that there is an ultraspinning instability for singly spinning BHs beyond $d \geq 6$. This is expected as beyond $d \geq 6$, angular momenta are allowed to become arbitrarily large for singly spinning MP BHs.

4 Black Rings

4.1 Geometries and Phase Space

We have now considered the stability of extended black objects and spinning BHs and gained some intuition. It is natural to look at both extended and spinning black objects, the family of BRs. These turn out to have many similarities to both BSs and MP BHs but also show new features.

Another motivation to studying BRs is that it has been shown by [36, 35] that the ultraspinning instability for MP BHs in $d \geq 6$ has a BR as a plausible end point.

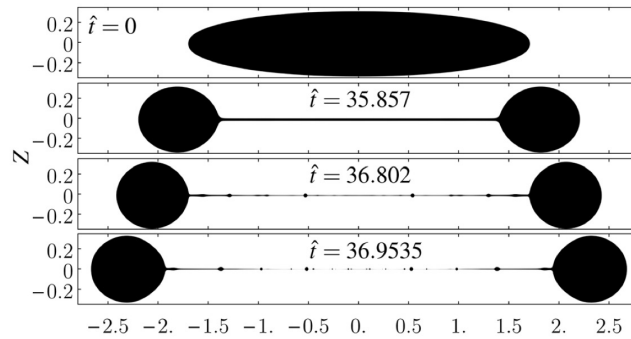


Figure 16: Simulation of apparent horizon (cross-section shown) dynamics for ultraspinning instability.

Image credit: [35]

A simulation performed in [35] can be seen in fig. 16 where a symmetric perturbation of a quickly rotating MP BH leads to it evolving into a set of concentric BRs. BRs are extended black objects, similar to BSs, with a horizon topology $S^{d-3} \times S^1$ where the S^1 is stabilised by angular momentum rather than topology. This is possible in any dimension $d \geq 5$ where the gravitational field falls off as quickly or more quickly than the centrifugal barrier.

A BR solution in $d = 5$ was first found by [9] and can be expressed as

$$ds^2 = -\frac{F(y)}{F(x)} \left(dt - CR \frac{1+y}{F(y)} d\psi \right)^2 + \frac{R^2}{(x-y)^2} F(x) \left(-\frac{G(y)}{F(y)} d\psi^2 - \frac{dy^2}{G(y)} + \frac{dx^2}{G(x)} + \frac{G(x)}{F(x)} d\phi^2 \right) \quad (4.1)$$

where $F(z) = 1 + \lambda z$, $G(z) = (1 - z^2)(1 + \nu z)$ and $C = \sqrt{\lambda(\lambda - \nu) \frac{1+\lambda}{1-\lambda}}$. We also have $\lambda = \frac{2\nu}{1+\nu^2}$, $0 \leq \nu < 1$ so the solution is specified by two parameters (ν, R) . Here R sets the overall scale and ν specifies the relative size of the S^2 and S^1 radii. The limit $\nu = 1$ corresponds to a singly spinning MP BH which has a ring singularity but an S^3 horizon. We can now introduce a dimensionless variable for area $a_H \sim \frac{A}{(GM)^{(d-2)/(d-3)}}$ and find the dimensionless area and momentum in terms of ν as

$$a_H = 2\sqrt{\nu(1-\nu)}, j = \sqrt{\frac{(1+\nu)^3}{8\nu}} \quad (4.2)$$

which has a cusp at $\nu = 1/2$. There are two branches of BRs for $\nu \leq 1/2$ and $\nu \geq 1/2$. The $\nu \geq 1/2$ branch has an extremal limit at $j = 1$ with $a_H = 0$ and only the $\nu \leq 1/2$ remains up to arbitrarily high angular momenta.

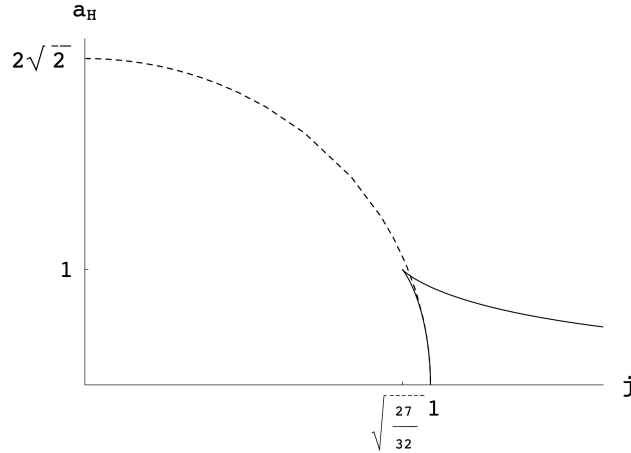


Figure 17: Phase space of BRs (solid) and singly spinning MP BHs (dashed) in $d = 5$.
Image credit: [8]

This is summarised in fig. 17 where the lower branch of the BRs corresponds to $\nu \leq 1/2$ and the upper branch to $\nu \geq 1/2$. Also shown is the singly spinning MP BH in $d = 5$ which experiences an extremal limit at the same angular momentum as the lower branch of BRs. The $\nu \geq 1/2$ branch is usually called *fat BRs* and the $\nu \leq 1/2$ branch are *thin BRs*.

For this metric, we only have angular momentum in the S^1 direction but, as we are already familiar with, in $d = 5$ there can be multiple independent angular momenta. The doubly spinning $d = 5$ BR has non-zero angular momentum in the S^2 directions. Its metric can be found in [41]. The only qualitative difference to the singly spinning BR is an additional parameter for the second angular momentum.

This phase space is shown in fig. 18 where we note regions where only MP BHs or only thin BRs are allowed as well as regions where all three types of spinning objects are allowed.

Rapidly spinning MP BHs can be formed from the collapse of a star and hence the BRs in the overlap region with the BHs should be able to form. This leaves the question whether thin BRs with angular momenta beyond what is allowed for BHs in $d = 5$ can be produced by a physical process. In [42], it was shown that for a sufficiently large number of particles in a high energy collision, BRs should be able to be formed in any

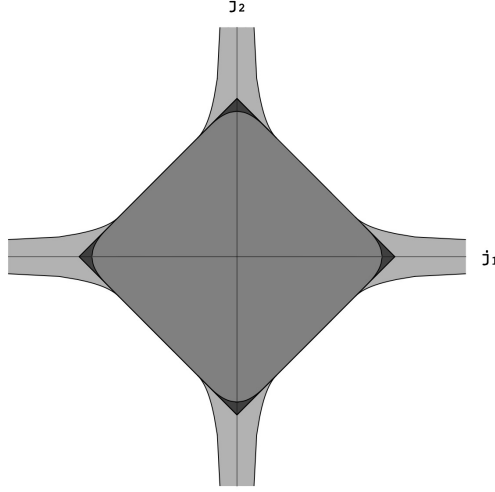


Figure 18: Phase space of BRs and MP BHs in $d = 5$. The central square allows MP BHs while the dark corners allow fat and thin BRs as well and the light grey regions allow thin BRs only.

Image credit: [8]

$d \geq 5$.

While no exact solutions are known for BRs in $d \geq 6$, one approach for this constructing approximate solutions is blackfolds. In the approach described by [43], BBs are wrapped around compact manifolds to generate more complex extended solutions. These can be solved using dynamics of an effective fluid. [44] Apart from blackfolds, there are also numerical methods for constructing BR solutions in $d \geq 6$. [45] We will now introduce methods for probing the stability of fat and thin BRs.

4.2 Stability of Black Rings

Turning Points

In fig. 17, we can see that for any given j , fat BRs have a lower entropy than MP BHs and thin BRs. This allows us to conjecture that fat BRs may be unstable and decay into MP BHs or thin BRs. Further, very thin BRs resemble boosted BSs so we may expect them to be unstable under a Gregory-Laflamme instability. To gather more evidence for this, we consider the method of turning points to probe for local thermodynamic stability of BRs rather than just the overall entropy difference between the branches.

To do this, we follow the formalism laid out in [46]. We assume there exists a smooth function S , called the Massieu function, which is equivalent to the entropy in our case, that the system tends to maximise. We define a space $\mathcal{M}$ of possible states of the system and let X be a point in this space, corresponding to a particular metric. For each point, there is also a set of thermodynamic variables $\{\mu_i\}$. In our case, these variables correspond to the mass and angular momentum on each branch of BRs. Small changes in X are metric perturbations away from the background BR metric. Equilibrium states are now defined as extrema of S under small changes dX that keep the $\{\mu_i\}$ constant $d\mu_i = 0$. We know the family of BRs are such equilibria for a given set of $\{\mu_i\}$ as they form stationary solutions of the EE. This allows us to write $dS = \beta_i d\mu^i$ at equilibrium points where β_i are uniquely defined for linearly independent $d\mu^i$. For equilibrium states, we can define $S_{eq} = S(\mu^i)$.

Now $\beta_i = \frac{\partial S_{eq}}{\partial \mu^i}$ gives the equation of state of the system. We can extend this function off equilibrium by defining

$$\hat{S} = \hat{S}(X^\rho; \mu^i) \quad (4.3)$$

where μ_i specifies the original equilibrium state and X^ρ parametrises the small perturbation away from it. To probe for stability of the BRs, we want to analyse the eigenvalues of the Hessian matrix $\partial_\rho \partial_\sigma \hat{S}$ evaluated at the equilibrium position. If any of these eigenvalues are positive, there is evidence for a local thermodynamic instability.

It can be shown that for any given β_a , we have

$$\frac{\partial \beta_a}{\partial \mu^a} = \partial_a^2 \hat{S} - \sum_\rho \frac{(\partial_\rho \hat{\beta}_a)^2}{\lambda_\rho} \quad (4.4)$$

evaluated at equilibrium. Here, $\hat{\beta}_i = \frac{\partial \hat{S}}{\partial \mu^i}$ and a is not summed over. This means that, as long as $\partial_a^2 \hat{S}$ is finite and $\partial_\rho \hat{\beta}_a$ is non-zero and finite, if the LHS of the equation diverges, we can attribute this to a change in sign of at least one of the λ_ρ and hence a change in stability. When β_a is plotted against μ^a , this corresponds to a turning point with a vertical tangent. There are cases, such as bifurcations in the phase diagram, where $\partial_\rho \hat{\beta}_a$ vanishes but these will not be relevant here. The reader may be directed to [46] for a more detailed description.

For BRs we now have $\{\mu^i\} = \{M, J\}$ and $\{\beta_i\} = \{\beta, \omega\}$ where $\beta = 1/T$ and $\omega = -\Omega/T$ where T is the black hole temperature and Ω is the event horizon angular velocity. We now have to plot β at fixed J and ω at fixed M to find the turning points.

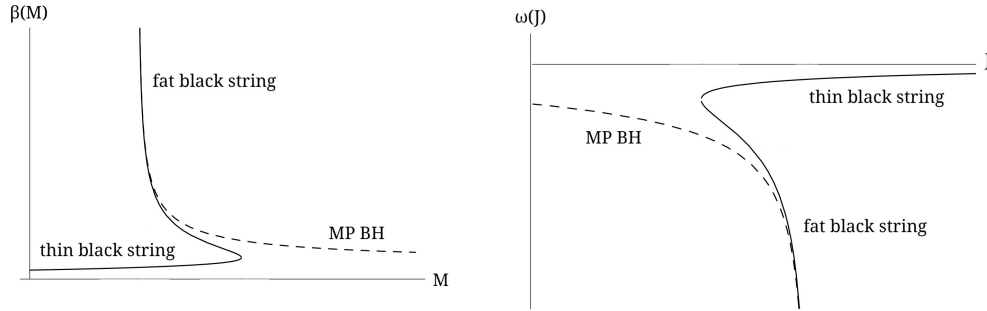


Figure 19: $\beta(M)$ at constant J on the left. $\omega(J)$ at constant M on the right.

Image credit: [46]

This is shown in fig. 19 where we can see there is such a turning point at $\nu = 1/2$. Hence there is a change in stability between the fat BR and thin BR branches. While this does not give us evidence that either fat or thin BRs are unstable, it means that they cannot both be stable. Based on the earlier argument that thin BRs have a higher entropy at the equal angular momenta, this is an indication that thin BRs are more stable than fat BRs.

Penrose Inequalities

Both thermodynamic and perturbation arguments often require a large amount of complexity in their computations. A simpler but powerful method is presented in [47] that uses a version of the Penrose inequality to construct a proof for the existence of instabilities. This can be used to show the onset of the ultraspinning instability for a MP BH and the Gregory-Laflamme instability of the BS however we are interested in its application to BRs.

In $d = 4$, the well-known Penrose inequality is

$$A_{min} \leq 16\pi E^2 \quad (4.5)$$

where E is the ADM energy of asymptotically flat initial data and A_{min} is the greatest lower bound for the area of a surface enclosing the apparent horizon. Crucially, this inequality assumes that the spacetime settles down to a Kerr solution. Hence, if we can replicate this in $d \geq 5$, we have a sufficient but not necessary condition for instability.

For rotating black objects, including BRs, the Penrose inequality is

$$A_{app} \leq A_{EH}(E, J_i) \quad (4.6)$$

where A_{app} is the area of the apparent horizon and $A_{EH}(M, J_i)$ is the area of the event horizon for the particular black object with mass M and angular momenta J_i . To show that an instability exists, we need to construct initial data that violated this inequality. This is done by considering initial data for the unperturbed BS $(\Sigma, \bar{h}_{ab}, \bar{K}_{ab})$ and applying a conformal rescaling

$$h_{ab} = \Psi(\lambda)^{4/(d-3)} \bar{h}_{ab}, K_{ab} = \Psi(\lambda)^{-2} \bar{K}_{ab} \quad (4.7)$$

where λ parametrises the rescaling with $\Psi(0) = 1$. To be valid initial data, Ψ automatically upholds the momentum constraint but has to satisfy an extra Hamiltonian constraint

$$\bar{\nabla}^2 \Psi - \frac{d-3}{4(d-2)} \bar{R}(\Psi - \Psi^{-3-4/(d-3)}) \quad (4.8)$$

where $\bar{\nabla}$ and $\bar{R}$ are the covariant derivative and Ricci scalar of $\bar{h}_{ab}$. When expanding to second order around $\lambda = 0$, this gives

$$\bar{\nabla}^2 \dot{\Psi} - \bar{R} \dot{\Psi} = 0, \bar{\nabla}^2 \ddot{\Psi} - \bar{R} \ddot{\Psi} = -\frac{3d-5}{d-3} \bar{R} \dot{\Psi}^2 \quad (4.9)$$

where $\dot{} = d/d\lambda$.

The ADM mass is modified to $M(\lambda)$ whereas the angular momentum $J(\lambda) = J$ is not for a radial perturbation.

It can now be shown that the Penrose inequality is equivalent to

$$Q \geq 0, Q = \ddot{M}(0) - \frac{1}{4} T \ddot{A}_{app}(0) - \frac{1}{T c_J} \dot{M}(0)^2 \quad (4.10)$$

where $c_J = \left(\frac{\partial M}{\partial T}\right)_J$ is the heat capacity at constant angular momentum. This changes the problem to finding a function $\Psi(\lambda)$ for which $Q < 0$. We notice that c_J is very similar to the inverse of $\left(\frac{\partial \beta}{\partial M}\right)_J$ considered in the turning points approach, up to a factor of $-T^2$. Hence c_J changes sign between the two branches and the final term contributes positively to Q for thin BRs and negatively for fat BRs.

Using the Hamiltonian constraint with the necessary boundary conditions allows finding a Ψ for which a rescaled $\bar{Q} \sim Q$ can be plotted against ν in fig. 20. This clearly shows that we have an instability for $\nu > 1/2$ which corresponds to the branch of fat BRs however is not able to make any statements about thin BRs. This confirms our previous thermodynamic argument arguing that at least one of the branches is unstable and our suspicion that the unstable branch is fat BRs. Q diverges at $\nu = 1/2$ as c_J passes through zero.

Applying the Penrose inequality to doubly spinning BRs, it can be confirmed that doubly spinning fat BRs are also unstable. Hence all fat BRs are unstable.

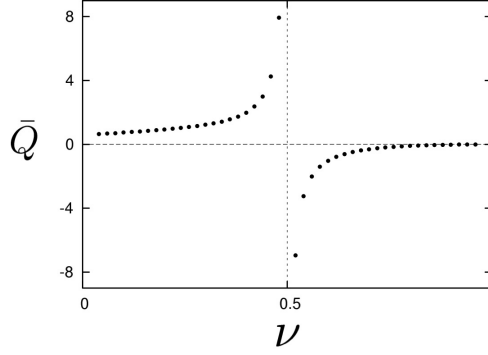


Figure 20: $\bar{Q} \sim Q$ against ν showing an instability for fat BRs.
Image credit: [47]

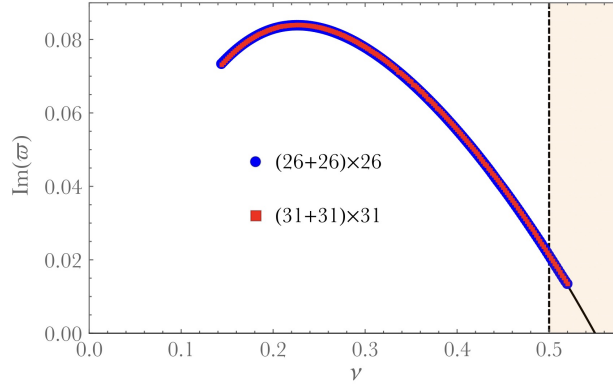


Figure 21: Imaginary part of frequency $\varpi \sim \omega$ against BR parameter ν for $m = 2$ perturbation mode.
Image credit: [48]

Perturbation Theory

The final tool that we apply to the analysis of BRs is perturbation theory. For non-axisymmetric perturbations, [48] have found a mode that becomes unstable on the fat branch but then continues across $\nu = 1/2$ to thin BRs. This instability is sometimes called an *elastic instability*.

This can be seen in fig. 21 as the imaginary part of the rescaled frequency $\varpi \sim \omega$ is positive over a range of values for the $m = 2$ mode.

The same non-axisymmetric instability is confirmed later by [49] in numerical evolution. It is found that this instability dominates the evolution for around $0.3 < \nu < 0.6$ whereas for very thin BRs with $\nu < 0.15$, a Gregory-Laflamme instability dominates. This should not come as a surprise as very thin BRs can be approximated by boosted BSs and should hence display similar dynamics.

Taking the results of thermodynamics, Penrose inequalities and perturbation theory together, this gives good evidence that all BRs are unstable. For fat black rings the radial mode dominates and drives them to spherical BHs as a plausible end point. The elastic and Gregory-Laflamme instabilities can be shown to end in fragmentation. [49]

Comparison of Approaches

We can take a step back to compare the tools we have introduced to probe the stability of BRs. Considering the global entropy difference between the two branches of BRs as well as the local turning points shows that thermodynamics can only give an indication that thin BRs are more stable here but does not make any statements about how many instabilities or of what type there are. The benefit of using turning points is that we can count how many eigenvalues of the entropy Hessian change sign and hence by how many unstable modes the branches differ from each other. Considering the validity of the thermodynamic argument for diagnosing classical instabilities, we can go back to the Gubser-Mitra conjecture, applicable to extended black objects with translational invariance. This argument should still be valid for very thin BRs which approximate boosted BSs but becomes a bit murky for less thin BRs and fat BRs. This emphasises that we may consider thermodynamic evidence but should not trust it alone.

Penrose inequalities provide a simpler method for analysing stability which does not require much computational complexity or knowledge of thermodynamic quantities. This method is powerful as it demonstrates that fat BRs are unstable but does not make any statements about thin BRs. Perturbation theory and numerical evolution turns out to be the most powerful method here, showing a range of instabilities that each dominate a certain part of the phase space.

By solving the simplified equations in the limit $d \rightarrow \infty$ to order $O(1/\sqrt{d})$, [50] shows that this behaviour stays qualitatively the same in large dimensions.

4.3 Phase Diagram

Armed with the knowledge from MP BHs and BRs, we can now attempt to construct a phase diagram of rotating black objects in $d \geq 6$.

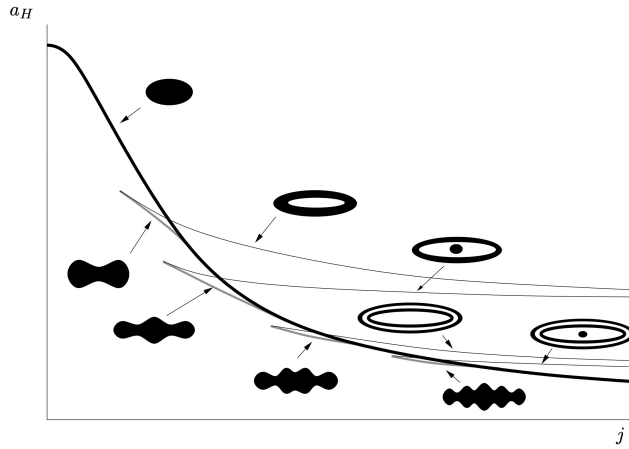


Figure 22: Proposed phase diagram for singly spinning black objects in $d \geq 6$.

Image credit: [51]

One such proposal by [51] is shown in fig. 22. This is an updated version of an earlier proposal by [52]. At low j , the only black object that can exist is a MP BH. As j is increased, this BH develops an ultraspinning instability causing the horizon to pinch in the middle which can evolve it into a BR, violating cosmic censorship when the horizon topology changes. For higher angular momentum, more ripples can develop and the horizon can pinch at multiple points which causes concentric rings to form, as seen in fig. 16. This allows the formation of more exotic objects such as di-rings and black saturns, a MP BH surrounded by one or more BRs.

An issue with this diagram is that it only considers axisymmetric perturbations to singly spinning BHs which overall form a measure zero subset of all BHs in higher dimensions. Adding other types of BHs can qualitatively change the diagram by adding other viable decay mechanisms and end points that we do not yet understand.

Completing a full phase diagram for black objects in any given dimension remains a difficult task, partly due to the requirement of classifying all possible BHs in higher dimensions.

Even if not all solutions in higher dimensions have been classified or analysed, the current understanding allows us to make an interesting observation. It appears that from all rotating BHs studied so far, only MP BHs can be stable. In $d = 4$, the only stationary BH is Kerr and hence eventually all BH spacetimes settle down to this by the no-hair theorem. In $d \geq 5$, there may be a comparable, weaker property that the end point to any BH spacetime is *generically* a MP BH. This would constitute a *dynamical* no-hair theorem although proving it presents a considerable challenge. While this still allows for the existence of other finely tuned, stationary solutions, it would imply that in reality all BHs eventually settle down to MP BHs.

As $d \rightarrow \infty$, we have seen that gravity dramatically simplifies. This does not only come with less complex equations but also more constrained dynamics. Studying the quasinormal mode spectrum of a non-rotating spherical BH has revealed that the spectrum tends to a limit where the late-time state has the same few frequencies present. This has carried through to the behaviour of rotating BHs where the spectrum of stable and unstable modes also approaches a limit. A limiting behaviour as $d \rightarrow \infty$ has also been observed for BSs, BBs and BRs. This limit has been studied by [53] who noted that as $d \rightarrow \infty$, the perturbation equations explicitly reduce to the theory of a fluid living on a membrane. This could be an indication that the phase diagram also simplifies as d gets very large.

So far we have only taken into account asymptotically flat rotating BHs in pure general relativity. Considering Einstein-Maxwell theory, we are able to compare this to charged BHs. For BRs in the large d limit, [54] find that the Gregory-Laflamme instability for thin BRs, while still existent, is weakened. An even more interesting result is reached when also including a Chern-Simons term. In this case, [55] have been able to construct a supersymmetric BR. As this BR saturates a BPS bound, it is a minimum energy state with decay pathways forbidden by supersymmetry and hence expected to be stable. This qualitatively changes the phase diagram if such a BR can be formed dynamically.

5 Conclusion and Outlook

Considering BSs and BBs, MP BHs and BRs not only reveals a richer structure of available BH solutions in $d \geq 5$ but also more interesting dynamics, including instabilities for generic initial data. Due to similarities with fluid dynamics, we have been able to gain intuition on the mechanisms of instabilities. These all appear to be manifestations of the same underlying competition between gravity, surface tension and the centrifugal force.

To show this mathematically, the powerful tools available in $d = 4$ cannot be trivially extended to $d \geq 5$ and instead, different methods, such as thermodynamics or Penrose inequalities, have to be introduced to diagnose instabilities. Each of the tools has its own issues and can be blind to or misdiagnose certain instabilities. This makes probing dynamic stability more complicated as a unified, robust framework is not yet available. Proposing conjectures about the more generic dynamics of BHs, such as it is done in [11] and [19] gives possible future avenues to developing such a framework. For all black objects we have analysed, we had many tools ready to diagnose the *onset* of instabilities but have always had to resort to numerical methods to track the subsequent evolution. This means that anything the numerical relativity code is sensitive to

will directly impact the accuracy of non-perturbative statements.

Therefore an unclear topic is that of the end point of any given instability. Although we have good evidence that black objects undergo phase transitions whereby the topology of the event horizon changes, there is no rigorous proof available. Furthermore, quantum gravity corrections become relevant at the level of the pinch-off and the classical description is not expected to be trustworthy. [18]

If we accept the proposed end points of instabilities that lead to topology changes in the event horizon, cosmic censorship is violated and the theory loses classical predictability. This is a big problem for physics as our model of gravity cannot be seen as a good classical theory where the whole spacetime can be predicted from a reasonable initial data slice. This loss of predictability is expected to be solved by quantum gravity theories that can make sense of the singularity without just cutting it out.

Studying a larger class of BHs with less symmetry could give an indication of how generic violations of cosmic censorship may be and how they might differ in higher dimensions. It is expected that these BHs display similar instabilities as the ones analysed, however as the vast majority of BHs in the limit $d \rightarrow \infty$ have not been classified, this is difficult to do, even qualitatively. Interestingly, in exactly this limit, where the dynamics simplify to an effective fluid description, we are closest to a unified framework.

If it is confirmed that cosmic censorship is generically violated for any $d \geq 5$ then $d = 4$ appears to be special as it is the lowest dimension for which asymptotically flat BHs exist but also the highest dimension for which general relativity classically predictable.

Finally, one of our motivations for studying gravity in $d \geq 5$ was gauge-gravity duality, learning about conformal field theories that map to unstable BHs. For this reason, it is of interest to investigate whether instabilities that appear for asymptotically flat BHs are also as ubiquitous in asymptotically AdS BHs and whether they cause topology changes through similar mechanisms. As a corollary to the work of [56], asymptotically AdS BHs, additionally to new instabilities, may experience an enhancement of the asymptotically flat instabilities as the timelike boundary of AdS space acts like a mirror, reflecting GWs and preventing perturbations from being dissipated. This tells us that these conformal field theories are not static but rather describe non-equilibrium processes such as thermalisation or phase transitions.

Higher-dimensional gravity challenges our four-dimensional assumptions about what gravity is fundamentally. We have seen that BHs are not unique, not stable and predictability appears to be lost. While we have established that gravity is *different* in higher dimensions, the *nature* of it remains open to questions and conjectures.

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